

TABLE 3
MAJOR SEMICONDUCTOR MANUFACTURING PROJECTS IN EAST AND SOUTHEAST ASIA

Company	Location	Investment Amount	Project Details	Expected Completion Date
TSMC (Taiwan Semiconductor Manufacturing Company)	Kumamoto, Japan	US\$8.6 billion	Japan Advanced Semiconductor Manufacturing, Inc. (JASM), a joint venture between TSMC (70%), Sony Semiconductor Solutions (20%), and Denso (10%), established Fab 23, producing 12-, 22-, and 28-nanometer processes.	Operational since December 2024
	Kumamoto, Japan	US\$13.9 billion	A second factory adjacent to Fab 23, currently under construction, will produce 6-nanometer and 12-nanometer processes.	Not specified
	Nanjing, Jiangsu, China	Not specified	Fab 16, a 300mm wafer fabrication facility operated by TSMC Nanjing Company Limited.	Operational
Powerchip Semiconductor Manufacturing Corporation (PSMC)	Miyagi Prefecture, Japan	Not specified	Joint venture with SBI Holdings to build a 12-inch (30.48cm) wafer fabrication facility at the Second Northern Sendai Central Industrial Park.	Not specified
Onsemi	Bucheon, South Korea	Not specified	Manufacturing facility focusing on 150mm and 200mm wafer production.	Operational
	Leshan, China	Not specified	Manufacturing facility.	Operational
	Shenzhen, China	Not specified	Manufacturing facility.	Operational
	Suzhou, China	Not specified	Manufacturing facility.	Operational
	Thuan An, Binh Duong, Vietnam	Not specified	Manufacturing facility.	Operational
	Bien Hoa, Dong Nai, Vietnam	Not specified	Manufacturing facility.	Operational
Nexchip	Hefei, China	Not specified	Joint venture between PSMC and Hefei City Construction and Investment Holding Group, established in 2015.	Operational
STMicroelectronics and Sanan Optoelectronics	Chongqing, China	US\$3.2 billion	Joint venture to establish an 8-inch (20.32cm) silicon carbide (SiC) device manufacturing facility. The project commenced production in February 2025, marking a significant advancement in SiC wafer manufacturing for automotive and industrial applications.	Operational since February 2025
TSMC	Hsinchu, Taiwan	Not specified	Fab 12A and Fab 12B, 300mm wafer fabrication facilities with multiple operational phases and ongoing expansions.	Operational and expanding
	Shanhua District, Tainan, Taiwan	Not specified	Fab 14 and Fab 18, 300mm wafer fabrication facilities with multiple operational phases and ongoing expansions.	Operational and expanding
	Taichung, Taiwan	Not specified	Fab 15, a 300mm wafer fabrication facility with multiple operational phases and planned expansions.	Operational and expanding
	Kaohsiung, Taiwan	Not specified	Fab 22, a 300mm wafer fabrication facility with planned multiple phases; Phase 1 is under construction.	Phase 1 under construction

tariffs. The Western world is bracing for a seismic shift in semiconductor policy.

Meanwhile, political and economic turbulence in Germany and France—Europe's largest economies—is undermining the EU's semiconductor ambitions. The Russia-Ukraine conflict has driven energy costs to record highs, disrupting European industry. Even if a ceasefire emerges under a Trump administration, the economic fallout will persist.

Germany's reliance on Russian natural gas has slowed its economy,

while France grapples with rising deficits, tax hikes, and spending cuts. The surge of far-right conservatism in both nations adds further uncertainty, stalling industrial projects. As a result, Germany's semiconductor ambitions are faltering, and France's chip plans remain in limbo.

According to *DIGITIMES Asia*, the current Trump presidency could further worsen these challenges, adding geopolitical and economic uncertainty and making it harder for Europe to regain its semiconductor competitiveness.

A *Deloitte* report highlights that Europe consumes twice as many chips as it manufactures, making it highly dependent on Asian suppliers. The region's reliance on East Asian chip imports has already caused disruptions, especially in industries like automotive, aerospace, and consumer electronics. The EU and the US efforts to localise chip production face two major challenges:

1. High costs. Building fabs in Europe or the US is significantly more expensive than in Asia.

2. Longer supply chains.